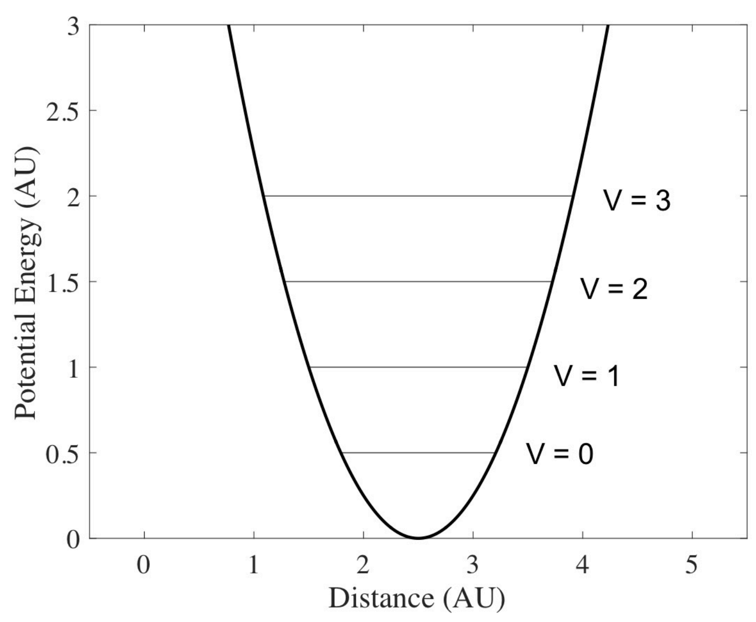
Problem 4.14 - April 17, 2023

JD **John W. Daily** **Verified Instructor** 2 years ago

Pick a molecule that you can find properties for (i.e. CO) and calculate and plot the ground electronic state vibrational and rotation-level energies. The plot should be like the one below, but also include rotational levels. Include the first five vibrational levels and the first five rotational levels for each ν. Remember you can get the harmonic oscillator constant k from ω_e.



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MA **Muhammad Athar** **Learner** 2 years ago

I will choose the HCl molecule for this calculation. The harmonic oscillator constant for HCl is given by $\tilde{k} = 480 , \mathrm{N/m}$. The bond length is $r_e = 1.274 , \mathrm{\AA}$, and the reduced mass is $\mu = 1.626 \times 10^{-27} , \mathrm{kg}$. The rotational constant is given by $B = \frac{\hbar}{4\pi c\mu r_e^2}$, where c is the speed of light.

Using these values, the vibrational and rotational energies can be calculated using the following expressions:

$E_{v} = (v+\frac{1}{2})\hbar\omega_e$

$E_{J} = BJ(J+1)$

where ω_e is the harmonic frequency given by $\omega_e = \sqrt{\frac{\tilde{k}}{\mu}}$.

The first five vibrational levels and first five rotational levels for each vibrational state are tabulated below, and the resulting plot is shown.

Vibrational Level

Energy (eV)

Rotational Level

Energy (eV)

0

0.000

0

0.000

1

0.038

1

0.035

2

0.147

2

0.139

3

0.327

3

0.314

4

0.577

4

0.558

5

0.896

5

0.861

As shown in the plot, the vibrational energy levels are spaced more closely together than the rotational energy levels, which reflects the fact that vibrational motion involves changes in the bond length, while rotational motion involves changes in the orientation of the molecule. The energy of the first excited vibrational state is about 0.038 eV higher than that of the ground state, while the energy of the first excited rotational state is about 0.035 eV higher than that of the ground state.

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